

BK Birla Institute of Engineering & Technology, Pilani
45 Days Summer Training Program
Subject – Data Science with AIML

Date : 17/06/2026

Topic – Looping, Python function, OOPS, Exception handling, File handling

Assignment 4:

1. Write a program that takes a list of numbers and prints all prime numbers in the list using loops. Do not use any built-in prime-checking functions.
2. Write a program that generates a multiplication table (1–10) in matrix format using nested loops.
3. Write a user-defined function that accepts a list of integers and returns:
 - Count of even numbers
 - Sum of odd numbers
 - Use loops and conditional statements inside the function.
4. Create a function that calculates simple interest.
Use default arguments for rate and time. Demonstrate calling the function using both positional and keyword arguments.
5. Create a Student class with attributes: name, marks. Add a method to calculate grade based on marks.
6. Create at least two objects and display their grades.
7. Create a BankAccount class with private attributes for balance. Provide methods to deposit and withdraw money. Ensure balance cannot be accessed directly.
8. Write a program that takes two numbers from the user and performs division.
 - Handle the following exceptions:
 - Division by zero

- Invalid input type

9. Write a program that:

- Creates a file
- Writes student data (name and marks)
- Reads and displays the content

10. Write a program that reads numbers from a file and calculates:

- Total
- Average
- Handle the case if the file does not exist.

Note: These Projects are meant strictly for your own learning and practice. Students are advised **not to copy answers from any website or online platform.**

Assignment submission form link -

<https://forms.gle/4TLL4p6kSrvDZFLL7>

